

SHOUCHEN ZHOU

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🎓 EDUCATION

Tsinghua University — Master of Science 2025.09 – Present

Major: Mathematics **GPA:** 3.9/4

ShanghaiTech University — Bachelor of Engineering 2021.09 – 2025.07

Major: Computer Science **GPA:** 3.91/4 **Major GPA:** 4/4 **Rank:** 1/178

🧑‍🔬 RESEARCH EXPERIENCE

Tsinghua University — Advisor: Prof. Chenglong Bao 2025.09 – Present

VRVC Lab, ShanghaiTech University — Advisor: Prof. Jingyi Yu 2022.08 – 2025.07

Conducted research on cryo-electron microscopy (cryo-EM) *ab initio* reconstruction, focusing on recovering 3D molecular structures from large collections of noisy and unposed 2D particle images. Contributed to CryoFastAR, a feed-forward multi-view reconstruction framework that estimates particle poses and reconstructs 3D structures without expensive iterative pose search, substantially improving reconstruction efficiency while maintaining competitive reconstruction quality.

Publication:

- **CryoFastAR: Fast Cryo-EM *Ab initio* Reconstruction Made Easy**

Jiakai Zhang, **Shouchen Zhou**, Haizhao Dai, Xinhang Liu, Peihao Wang, Zhiwen Fan, Yuan Pei, Jingyi Yu

In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025. [project page](#)

⚙️ TEACHING EXPERIENCE

Led weekly recitation sessions; graded coursework and exams, including programming assignments when applicable; and provided academic support and feedback to students.

Tsinghua University 2026.09 – Present

Teaching Assistant for **Probability and Statistics** (Fall 2026).

ShanghaiTech University 2023.02 – 2025.06

Teaching Assistant for five courses, including **Linear Algebra**, **Probability and Statistics**, **Algorithms and Data Structures**, and **Introduction to Machine Learning**.

💖 AWARDS

Shanghai Outstanding Graduate 2025.06

Outstanding Graduate, ShanghaiTech University 2025.06

Shanghai Scholarship 2024.12

Outstanding Student, ShanghaiTech University (Academic Year 2023) 2024.12

Outstanding Student, ShanghaiTech University (Academic Year 2022) 2023.12

Outstanding Student, ShanghaiTech University (Academic Year 2021) 2022.12

💻 SKILLS

- Programming Language: Python, C, C++, MATLAB, $\text{\LaTeX}$
- Language ability: CET-4 : 570, CET-6 : 551
- Platform: Linux(Ubuntu), Windows